

Stage 7R Certification and Persistence Boundary Specification

Scientific Results, Derived Indices, Search Workspaces, and Verification Certificates

mdstats

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Objective

Stage 7R consolidates the implemented topology, ring, symmetry, and strength stack before Euclidean embedding. It enforces the architectural distinction

scientific result $\neq$ derived index $\neq$ search workspace $\neq$ verification certificate

Runtime/API target:

mdstats 0.19.19a0

No primitive-ring classification, periodic automorphism, symmetry-group order, or bounded strength theorem is changed.

Boundary table

Category	Persistent?	Reconstructible?	Example
Scientific result	yes	not assumed	PeriodicNetSymmetry, RingStrengthResult
Derived index	optional	yes	PrimitiveRingSymmetryIndex, PrimitiveRingIndex
Search workspace	no	yes	RingStrengthSearchWorkspace
Positive certificate	yes	independently checkable	RingStrengthWitness
Negative bounded certificate	compact result + deterministic replay	yes	STRONG_IN_DOMAIN

Implemented refactors

Core net symmetry

PeriodicNetSymmetry schema v3 stores only:

- source view/topology identities;
- normalized automorphism representatives;
- multiplication table;
- translation cocycle;
- inverse and identity indices;
- vertex orbits; and
- edge orbits.

It contains no primitive-ring keys, actions, or ring-search provenance.

Primitive-ring symmetry index

PrimitiveRingSymmetryIndex binds one exact group to one exact ring catalog and uses integer target positions in the action table. It records ring-catalog completeness metadata and validates the induced action homomorphism.

Strength result versus workspace

RingStrengthResult schema v2 stores:

```
target_placement
domain
resources
status
diagnostics
candidate_set_digest
witness | None
```

It no longer stores every candidate placement. The deterministic transient RingStrengthSearchWorkspace stores the exhaustive candidate tuple when a caller explicitly asks for it.

For a candidate set $\mathcal{P}$, the persistent record stores

$$d_{\mathcal{P}} = \text{SHA256}(\text{canonical_json}(\mathcal{P})).$$

Independent strength verification

RingStrengthResult.verify(index) performs two checks:

1. for WEAK_CERTIFIED, recompute the physical lifted-edge parity of the target and witness components;
2. deterministically rebuild the declared finite domain, rerun the bounded cancellation solve, and require canonical result equality.

from_dict(..., verify=True) uses this path by default. Recomputing a payload digest is therefore insufficient to forge a scientific certificate.

GF(2) memory policy

The finite cancellation backend now declares conservative capacities:

```
FiniteRingCancellationResources(
    max_matrix_bits,
    max_provenance_bits,
)
```

For N candidates and E physical edge basis elements, the preflight estimates

$$B_{\text{matrix}} = NE, \quad B_{\text{provenance}} = N \min(N, E).$$

Exceeding either bound raises a resource error. The strength layer translates it to UNRESOLVED_TRUNCATED; it never reports a false negative strength result.

Shared barycentric placement

The exact rational equilibrium solve is extracted to `PeriodicBarycentricPlacement`. Symmetry discovery consumes that source-bound object instead of carrying an anonymous private coordinate tuple.

Fast view lookup

`PeriodicNetView` constructs transient immutable maps

atom index -> vertex position

`FrameworkEdgeKey` -> edge position

so repeated automorphism and ring-action lookups are $O(1)$ rather than linear searches.

Schema changes

Object	Previous	Current
<code>PeriodicNetSymmetry</code>	v2	v3
<code>PeriodicNetSymmetryDiscovery</code>	v1	v2
<code>RingStrengthResult</code>	v1	v2
<code>RingStrengthCatalog</code>	v1	v2
<code>PrimitiveRingSymmetryIndex</code>	absent	v1
<code>PeriodicBarycentricPlacement</code>	absent	v1

These are intentional alpha API changes. Legacy payloads are not silently upgraded because their ownership boundaries are ambiguous.

Measured Na-LTA effect

For the 96-operation Na-LTA symmetry and 82 primitive rings:

core <code>PeriodicNetSymmetry</code> JSON	~0.83 MB
<code>PrimitiveRingSymmetryIndex</code> JSON	~1.01 MB
combined discovery JSON	~1.84 MB

The former combined ring-heavy symmetry payload was approximately 9.17 MB.

For one depth-eight 8-ring strength calculation with 3,240 candidate placements:

persistent <code>RingStrengthResult</code> JSON	~4 KB
candidate workspace	transient/reconstructible

The earlier persistent result scaled approximately linearly with every complete candidate key and placement.

Deferred work

Stage 7R deliberately does not implement:

- symmetry-orbit-level strength transport;
- a sparse or two-pass GF(2) solver;
- a compact standalone negative row-space certificate;
- a Euclidean invariant Gram matrix;
- automatic disk/face construction; or
- natural tiling.

The symmetry-equivariant strength layer remains required before final face selection, but it does not block Stage 8A embedding.

Acceptance gate

The consolidation passes when:

1. core symmetry round trips without ring data;
2. ring action round trips only with the exact group and ring catalog;
3. strength payloads contain no candidate workspace;
4. forged weak witnesses fail independent verification;
5. GF(2) memory limits return unresolved status;
6. direct barycentric placement and discovery produce identical exact coordinates and symmetry groups;
7. all prior Stage-4–7 scientific regressions pass; and
8. package-wide tests reveal no import or schema regressions.